

The most significant issue was a **Failure in Rule Application** (44.8%), where models disregard basic geometric axioms, such as the spatial relationship between adjacent and opposite faces on a cube. This suggests an inability to link visual data with abstract principles. The second major deficit was a **Failure in Mental Simulation** (40.0%), indicating a lack of "spatial working memory" to track objects through transformations, as seen in Fold and Punch where state changes are consistently miscalculated. Finally, a **Failure in Holistic-Local Processing** (15.2%) was observed, where models cannot appropriately shift attention between an object’s overall structure and its local details, often being misled by superficial similarities while ignoring critical flaws. Detailed error analysis pipeline, definition of error categories and more discussion is presented in Appendix C.

5.2 DISENTANGLING PERCEPTUAL CONFOUNDERS

Table 5: Layout Ablation Study on Qwen2.5-VL-7B to Disentangle Perceptual Confounders.

Input Setting	Accuracy
Standard Merged Layout	26.1%
Separate Images Layout	24.9%

Although the preceding analysis categorizes the overwhelming majority of failures as reasoning deficits, confirming the absolute rigor of this conclusion requires eliminating potential perceptual bottlenecks. To ensure that our error analysis accurately captures genuine spatial reasoning deficits rather than superficial perceptual confounders—such as an inability to parse complex, multi-panel visual layouts—we conducted a layout ablation study. In our standard evaluation setting, the question and corresponding options are presented to the model as a single merged image. It is plausible that a model’s failure could stem from visual parsing issues rather than spatial reasoning limits.

To test this, we compared the standard merged-image setting against a separate-images setting, where the question and each option are provided to the model as independent image inputs. Evaluating Qwen2.5-VL-7B under these two conditions yielded near-identical accuracies: 26.1% for the standard merged layout and 24.9% for the separate images. If layout parsing were the primary bottleneck, isolating the images should have resulted in a clear performance improvement. The absence of such a gain strongly supports our qualitative findings that the dominant difficulty for VLMs lies in complex spatial-logical inference—such as maintaining 3D adjacencies and global consistency—rather than format parsing.

6 CONCLUSION AND LIMITATIONS

Conclusion and future direction. We introduced **Spatial-DISE** and Spatial-DISE-12K to comprehensively evaluate spatial reasoning in VLMs. Our findings reveal universal cognitive deficits, particularly in applying geometric rules and performing mental simulations. This work provides a necessary framework to guide future VLM development.

To advance spatial intelligence, future research must shift from mere visual perception to active, human-like reasoning. Key directions include: **(1)** closing the sim-to-real gap by transferring abstract cognitive concepts derived from synthetic settings; **(2)** evolving evaluations from isolated puzzles to interactive tasks (e.g., navigation and robot manipulation) ; and **(3)** adopting process-oriented assessments that require textual justifications or action plans, distinguishing true mental simulation from fragile heuristic matching

Limitations. Our error analysis relies on a hybrid LLM+human pipeline in which Doubao-1.6-thinking proposes an error type and explanation that a human annotator then verifies. While this reduces manual effort, the LLM’s initial label may bias the annotator, and analysing only 200 errors from four models means Table 4 should be read as qualitative trends rather than precise estimates. Additionally, Spatial-DISE-12K consists entirely of synthetic 3D scenes; despite consistent gains on real-world benchmarks, a sim-to-real gap may persist for naturalistic settings. Human quality control covers a stratified sample of 1,000 items, with programmatic single-solution checks applied to the remainder, as exhaustive manual verification of the full 12K is infeasible.

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LLM Usage Statement. We declare that large language models (LLMs) were used exclusively for language editing and stylistic improvements in this manuscript. They did not contribute to the conceptual, methodological, or experimental aspects of the work.

Ethics Statement. This work adheres to ethical research practices. Human performance data were collected from 54 participants. The study protocol was reviewed and approved by the Faculty of Science and Engineering Research Ethics Committee at the University of Liverpool (Project ID: 16623). Prior to participation, all individuals were informed of the study’s purpose and procedure. They provided digital informed consent before accessing the tasks and were explicitly notified of their right to withdraw at any time. No sensitive personal data was collected. All response data are anonymized and will be destroyed three months after the paper is accepted.

Reproducibility Statement. We provide comprehensive details to ensure full reproducibility. The complete dataset curation process, including the synthetic data generation pipeline, is detailed in Section 3.2 and Appendix A.4. This includes procedural algorithms (pseudocode) and specific implementation details for the five core 3D tasks. All evaluation settings, including benchmark models, baselines, and implementation details, are described in Section 4.1. Our human performance assessment methodology is thoroughly documented in Appendix B.1. The benchmark, dataset, and code will be made publicly available to facilitate direct comparison with our results.

REFERENCES

- Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, et al. Flamingo: A Visual Language Model for Few-Shot Learning. In *NeurIPS 2022*, 2022.
- Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, et al. Gemini: A Family of Highly Capable Multimodal Models. *arXiv:2312.11805*, 2025.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, et al. Qwen2.5-VL Technical Report. *arXiv:2502.13923*, 2025.
- G. K. Bennett, H. G. Seashore, and A. G. Wesman. *Differential Aptitude Tests*. Differential Aptitude Tests. 1947.
- GEORGE M. BODNER and ROLAND B. GUAY. The Purdue Visualization of Rotations Test. *The Chemical Educator*, 2:1–17, 1997. ISSN 1430-4171.
- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh, Leonidas Guibas, and Fei Xia. SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. In *CVPR2024*, 2024.
- Hardy Chen, Haoqin Tu, Fali Wang, Hui Liu, Xianfeng Tang, Xinya Du, Yuyin Zhou, and Cihang Xie. SFT or RL? An Early Investigation into Training R1-Like Reasoning Large Vision-Language Models. *arXiv:2504.11468*, 2025.
- Xi Chen, Xiao Wang, Lucas Beyer, Alexander Kolesnikov, et al. PaLI-3 Vision Language Models: Smaller, Faster, Stronger. *arXiv:2310.09199*, 2023.
- An-Chieh Cheng, Hongxu Yin, Yang Fu, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, and Sifei Liu. SpatialRGPT: Grounded Spatial Reasoning in Vision Language Models. In *NeurIPS 2024*, 2024.
- Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, et al. Kimi-VL Technical Report. *arXiv:2504.07491*, 2025.
- Haodong Duan, Xinyu Fang, Junming Yang, Xiangyu Zhao, Yuxuan Qiao, Mo Li, Amit Agarwal, Zhe Chen, Lin Chen, Yuan Liu, Yubo Ma, Hailong Sun, Yifan Zhang, Shiyin Lu, Tack Hwa Wong, Weiyun Wang, Peiheng Zhou, Xiaozhe Li, Chaoyou Fu, Junbo Cui, Xiaoyi Dong, Yuhang Zang, Pan Zhang, Jiaqi Wang, Dahua Lin, and Kai Chen. VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models. In *MM ’24: The 32nd ACM International Conference on Multimedia*, 2025.

- Ruth B. Ekstrom and Harry Horace Harman. *Manual for Kit of Factor-Referenced Cognitive Tests*, 1976. 1976.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: Multimodal Large Language Models Can See but Not Perceive. In *ECCV 2024*, 2024.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, et al. The Llama 3 Herd of Models. *arXiv:2407.21783*, 2024.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, et al. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. *arXiv:2501.12948*, 2025a.
- Dong Guo, Faming Wu, Feida Zhu, Fuxing Leng, Guang Shi, et al. Seed1.5-VL Technical Report. *arXiv:2505.07062*, 2025b.
- Wenyu Han, Siyuan Xiang, Chenhui Liu, Ruoyu Wang, and Chen Feng. SPARE3D: A Dataset for SPATial REasoning on Three-View Line Drawings. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- Mengdi Jia, Zekun Qi, Shaochen Zhang, Wenyao Zhang, Xinqiang Yu, Jiawei He, He Wang, and Li Yi. OmniSpatial: Towards Comprehensive Spatial Reasoning Benchmark for Vision Language Models. *arXiv:2506.03135*, 2025.
- Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s “up” with vision-language models? Investigating their struggle with spatial reasoning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9161–9175, 2023.
- Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models. *arXiv:2301.12597*, 2023.
- Liunian Harold Li, Pengchuan Zhang, Haotian Zhang, Jianwei Yang, Chunyuan Li, Yiwu Zhong, Lijuan Wang, Lu Yuan, Lei Zhang, Jenq-Neng Hwang, Kai-Wei Chang, and Jianfeng Gao. Grounded Language-Image Pre-training. In *CVPR 2022*, 2022.
- Yuan-Hong Liao, Rafid Mahmood, Sanja Fidler, and David Acuna. Reasoning Paths with Reference Objects Elicit Quantitative Spatial Reasoning in Large Vision-Language Models. In *EMNLP 2024*, 2024.
- Marcia C. Linn and Anne C. Petersen. Emergence and Characterization of Sex Differences in Spatial Ability: A Meta-Analysis. *Child Development*, 56:1479, 1985. ISSN 0009-3920.
- Fangyu Liu, Guy Emerson, and Nigel Collier. Visual Spatial Reasoning. *Transactions of the Association for Computational Linguistics*, 11:635–651, 2023. ISSN 2307-387X.
- Shiyin Lu, Yang Li, Qing-Guo Chen, Zhao Xu, Weihua Luo, Kaifu Zhang, and Han-Jia Ye. Ovis: Structural Embedding Alignment for Multimodal Large Language Model. *arXiv:2405.20797*, 2024.
- Wufei Ma, Haoyu Chen, Guofeng Zhang, Yu-Cheng Chou, Celso M de Melo, and Alan Yuille. 3DSRBench: A Comprehensive 3D Spatial Reasoning Benchmark. In *ICCV 2025*, 2025.
- P.H Maier. Spatial Geometry and Spatial Ability- How to Make Solid Geometry Solid. In *Proceedings of the Annual Meeting of the GDM*, 1996.
- Kun Ouyang, Yuanxin Liu, Haoning Wu, Yi Liu, Hao Zhou, Jie Zhou, Fandong Meng, and Xu Sun. SpaceR: Reinforcing MLLMs in Video Spatial Reasoning. *arXiv:2504.01805*, 2025.
- George J. Pallrand and Fred Seeber. Spatial ability and achievement in introductory physics. *Journal of Research in Science Teaching*, 21:507–516, 1984. ISSN 0022-4308.
- Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. LMM-R1: Empowering 3B LMMs with Strong Reasoning Abilities Through Two-Stage Rule-Based RL. *arXiv:2503.07536*, 2025.